Process Architecture and Process Profile.

1)

Student Application and Admission		
Name of Process:	Student Application and Admission	
Vision	To ensure a transparent, efficient and fair admission process that selects qualified students based on academic merit and required qualifications. @9ualfzte	
Process Owner:	Admissions Registrar	
Customer of process	• Prospective Students	**Expectation of Customer:** • Fair admission process • Clear communication • Quick response and decision
Outcome	Qualified students are officially admitted to the University	
Trigger	Student submits application	
First activity	Submit student application	
Last activity	Issue admission letter	
Interfaces inbound:	Student Recruitment	
Interfaces outbound:	Student Registration and Enrollment	
Required resources	• **Homan resources:** Admission officers, Academic evaluators, interview panell • **Information, documents,** know-how: Student applications, Academic certificates, policies and guidelines • **Work environment, materials, infrastructure:** Admission management system, online application system	
Process Performance Measures	• Application processing time • Number of qualified applicants admitted	

2)

Student Registration and Enrollment		
Name of Process:	Student Registration and Enrollment	
Vision	To accurately register admitted students and provide them with access to university systems, enabling them to begin their academic programs without delay.	
Process Owner:	Registrar	
Customer of process	Newly admitted students	**Expectation of Customer:** • Smooth registration process
Outcome	Fast access to university systems. Students are officially registered and enrolled in thedegree program.	
Trigger	Student accepts admission offer	
First activity	Accept admission offer	
Last activity	Provide VLE access and university ID	
Interfaces inbound	Student Application and Admission	
Interfaces outbound	Deliver Academic Programs and Services	
Required resources	• Human resources: Registration officers, IT support staff • Information and documents: Student admission records, registration guidelines. • Work environment, materials, infrastructure: Student information systems, virtual learning environment	
Process Performance Measures	• System access rate • Student registration completion rate • Registration processing time	

3)

Deliver Academic Programs and Services		
Name of Process:	Deliver Academic Programs and Services	
Vision	Provide high-quality academic education, learning resources, teaching support that help students achieve their educational goals.	
Process Owner:	Head of Department (HOD)	
Customer of process	Registered students	**Expectation of Customer:** • Quality teaching • Access to learning resources • Effective academic supports
Outcome	Students receive academic instruction and learning resources.	
Trigger	Semester begins	
First activity	Lecture hall arrangement	
Last activity	Facilitate group projects and presentation	
Interfaces inbound:	Student Registration and Enrollment	
Interfaces outbound:	Student Assessment and Evaluation ▶	
Required resources	• Human resources: Lecturers, Lab instructors, Academic coordinators • Information and documents: Course curriculum, Lecture materials, Academic guidelines • **Work environment, materials,** infrastructure: Classrooms and laboratories, VLE Platform, Library	
Process Performance Measures	• Course completion rate • Teaching quality evaluation • Student satisfaction	

4)

Student Assessment and Evaluation		
Name of Process:	Student Assessment and Evaluation	
Vision	Evaluate student academic performance fairly and accurately through structured assessments, examinations and project evaluations.	
Process Owner:	Examination Department and Academic Coordinator	
Customer of process	• Students • Academic departments	**Expectation of Customer:** • Fair evaluation • Accurate grading • Timely feedback
Outcome	Student performance is assessed and graded.	
Trigger	Completion of coursework or exams.	
First activity	Conduct continuous assessments	
Last activity	Evaluate assignments and projects.	
Interfaces inbound:	Deliver Academic Programs and Services	
Interfaces outbound:	Academic Results Processing	
Required resources:	• Human resources: Lecturers, Examination officers • **Information, documents:** Exam papers, Marking schemas, Assessment guidelines • **Work environments:** Examination halls, online exam systems	
Process Performance Measures	• Grading accuracy • Evaluation time • Student performance results	

5)

Student Support and Development		
Name of Process:	Student Support and Development	
Vision	To support student personal, academic and career development through counseling, extracurricular activities and career guidance programs.	
Process Owner:	Student Affairs Department	
Customer of process	• Students • Academic departments	**Expectation of Customer:** • Guidance and support • Career opportunities • Personal development
Outcome	Student gain support, skills and career opportunities.	
Trigger	Student requires support or development activities.	
First activity	Provide counseling and mentoring	
Last activity	Organize career fairs and workshops	
Interfaces inbound:	Deliver academic programs and Services	
Interfaces outbound:	Research and training	
Required resources:	• Human resources: Counselors, Career advisors, Student activity coordinators • Information, documents: Career development programs, Internship information • Work environment: Career platforms, counseling centers, Event halls	
Process Performance Measures	• Student participation rate • Internship placement rate • Student satisfaction	

6)

Research and Training		
Name of Process:	Research and Training	
Vision	Promote research, innovation and Industry exposure by engaging students in research projects and practical training opportunities.	
Process Owner:	Research Coordinator and Industry Coordinator	
Customer of process	• Students • Academic staff • Industry partners	**Expectation of Customer:** • Quality research opportunities • Practical industry exposure
Outcome	Students complete research projects and industrial training.	
Trigger	Start of research or internship requirement.	
First activity	Assign research supervisors & Internship coordinator	
Last activity	Complete final year research and training.	
Interfaces inbound:	Student Support and Development	
Interfaces outbound:	Graduation and Completion	
Required resources:	• Human **resources:** Research supervisors, Industry mentors • **Information, documents:** Research guidelines, Project documentation • Work **environments:** Research places, Industry partner organizations	
Process Performance Measures	• Research completion rate • Research publication • Internship success rate	

7)

Academic Results Processing		
Name of Process:	Academic Results Processing	
Vision	To process and publish academic results accurately and on time, ensuring transparency and reliability in student performance records.	
Process Owner:	Examination Department	
Customer of process	• Students • Academic departments	**Expectation of Customer:** • Accurate results • Timely publication
Outcome	Official semester results and transcripts are issued.	
Trigger	Completion of assessments and grading	
First activity	Grade assignment and exams.	
Last activity	Issue transcripts	
Interfaces inbound:	Student Assessment and Evaluation	
Interfaces outbound:	Graduation and Completion	
Required resources	• Human resources: Examination officers, Academic staff • Information, documents: Grade sheets, Examination records • Work environment: Examination portal	
Process Performance Measures	• Result processing time • Accuracy of results • Transcript issuance time	

8)

Graduation and Completion		
Name of Process:	Graduation and Completion	
Vision	To verify academic completion requirements and award degrees to eligible students in a timely and organized manner.	
Process Owner:	Registrar/Exam Division / Student Affairs Office	
Customer of process	• Graduating Students	Expectation of Customer: • Timely graduation • Official degree certification
Outcome	Students receive degrees and become university alumni.	
Trigger	Student completes all academic requirements	
First activity	Verify degree completion requirements.	
Last activity	Maintain alumni relationships.	
Interfaces inbound:	Academic Results Processing	
Interfaces outbound:	Alumni Relation	
Required resources	• Human resources: Academic administrators, Graduation committee • Information, Documents: Student academic records, Degree requirements • Work Environment: Graduation management system, Convocation Facilities.	
Process Performance Messures	• Graduation processing time • Degree verification accuracy • Alumni engagement rate	

Process Portfolio

Prioritizing - IT Undergraduate Journey at UWO

High	**Selection Focus** • Time Management & Planning • Deep Technical skill building • Critical Thinking & Problem Solving • Exam & Assignment Management	• Project Development Skills (Mini projects, Group work) • Communication & Presentation skills • Teamwork & Collaboration
Strategic Importance	• Participation in Extra-curricular clubs. • Miscellaneous Administrative Tasks.	• Hobby-based Learning (Gaming, editing, design etc) • Basic soft skills (Email writing)
Low	**Poor**	**Health** **Good**

Feasibility

- **Low**
- **Medium**
- **High**

Process Analysis

Quantitative Analysis

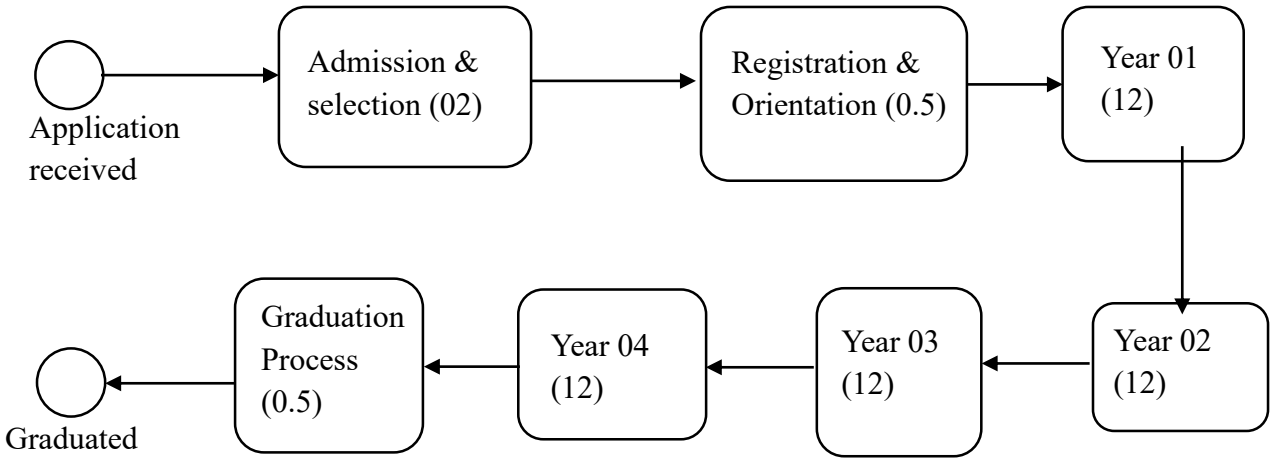
Flow Analysis

Process Stages & Time Duration

Process stage	Time (Month)
Admission & selection	02
Registration & Orientation	0.5
Year 01 - Academic Learning	12
Year 02 - Academic Learning	12
Year 03 - Academic Learning	12
Year 4.1 - Academic Learning	06
Year 4.2 - Internship	06
Graduation processing	0.5

Assumption:

- Project I, II also included in respect to year 02, 03 with academics
- Research also included in respect to year 4.1 and 4.2 with academics and internship



$$CT = \sum_{i=1}^{n} T_i$$

= 2 + 0.5 + 12 + 12 + 12 + 12 + 0.5

= 51 Months (4.25 years)

Process Analysis

Qualitative Analysis

Value Added Analysis and Waste Analysis

Process	Value Added	Business Value Added	Waste
Meet university requirements & criteria	Enabling students to progress through their academic programme successfully	Ensure compliance with university standards and academic regulations	Time is wasted due to repeated document submissions and unclear requirements
Study core IT subjects	Develop essential technical knowledge required for IT field	Supports the university in producing industry-ready graduates	Occur inefficiency when outdated or less relevant topics are included
Group projects	Improve teamwork, collaboration, and communication skills	Prepare students for collaborative work environments in organizations	Arise delays and inefficiencies due to poor coordination and unequal participation
Apply theories to real world	Transform theoretical knowledge into practical skills	Improve job readiness and real-world problem-solving capability	Occur waste when there are insufficient opportunities for practical exposure
Internship	Provide hands-on industry experience	Enhance employability and strengthen the university reputation	Waste time if tasks lack structure or proper supervision
Manage academic processes	Value the necessity of administration	Ensure proper coordination of academic activities	Waste time due to manual processes and repeated approvals
Manage extra activities	Develop soft skills and personal growth	Produce well-rounded graduates	Time conflicts with academic responsibilities reduce effectiveness
Become a well-rounded graduate	Represents the final value created through the entire process	Fulfill the university's objective of producing competent professionals	Overall inefficiencies across the process reduce the full potential outcome

Issue Registry

Issue 1: Delays in Academic Administrative Processes

Priority: 1

Description:

Students experience delays due to manual academic processes such as registration, approvals, and document handling.

Data and Assumptions:

Assume 1,000 students. Each student wastes 5 hours per semester due to delays.

Qualitative Impact:

Causes frustration, reduces productivity, and delays academic progress.

Quantitative Impact:

1,000 × 5 × 2 semesters = **10,000 hours wasted per year**

Issue 2: Inefficient Communication Channels

Priority: 1

Description:

Academic information is scattered across multiple platforms (LMS, emails), causing confusion and delays.

Data and Assumptions:

Assume each student wastes 3 instances × 30 minutes per semester.

Qualitative Impact:

Leads to missed deadlines, miscommunication, and reduced coordination.

Quantitative Impact:

1,000 × 3 × 0.5 × 2 = **3,000 hours wasted per year**

Issue 3: Limited Practical Exposure (Completed)

Priority: 1

Description:

Students have insufficient opportunities to apply theoretical knowledge in real-world scenarios.

Data and Assumptions:

Assume students lose 4 hours per week in ineffective learning over 30 weeks.

Qualitative Impact:

Reduces skill development and job readiness.

Quantitative Impact:

1,000 × 4 × 30 = 120,000 hours of ineffective learning per year

Issue 4: Poor Coordination in Group Projects

Priority: 2

Description:

Group projects suffer from unequal participation and lack of coordination among students.

Data and Assumptions:

Assume each student wastes 2 hours per project with 4 projects per year.

Qualitative Impact:

Creates conflicts, reduces teamwork effectiveness, and lowers output quality.

Quantitative Impact:

1,000 × 2 × 4 = 8,000 hours wasted per year

Issue 5: Outdated or Less Relevant Study Materials

Priority: 2

Description:

Students spend time learning outdated or less industry-relevant content.

Data and Assumptions:

Assume 10% of study time (5 hours per week) is ineffective over 30 weeks.

Qualitative Impact:

Reduces learning effectiveness and industry readiness.

Quantitative Impact:

$1{,}000 \times 5 \times 0.1 \times 30 = 15{,}000$ hours wasted per year

Issue 6: Unstructured Internship Experience (Completed)

Priority: 2

Description:

Some internships do not provide meaningful or structured learning opportunities.

Data and Assumptions:

Assume 50% of students waste 2 weeks (40 hours per week).

Qualitative Impact:

Limits skill development and reduces internship value.

Quantitative Impact:

$500 \times 80 =$ **40,000 hours wasted per year**

Issue 7: Limited Access to Tools and Software

Priority: 3

Description:

Students face difficulties accessing required tools and software for practical learning.

Data and Assumptions:

Assume each student loses 2 hours per month over 10 months.

Qualitative Impact:

Reduces practical learning efficiency and skill development.

Quantitative Impact:

1,000 × 2 × 10 = **20,000 hours wasted per year**

Issue 8: Time Conflicts due to Extra Activities

Priority: 3

Description:

Students struggle to balance extracurricular activities with academics.

Data and Assumptions:

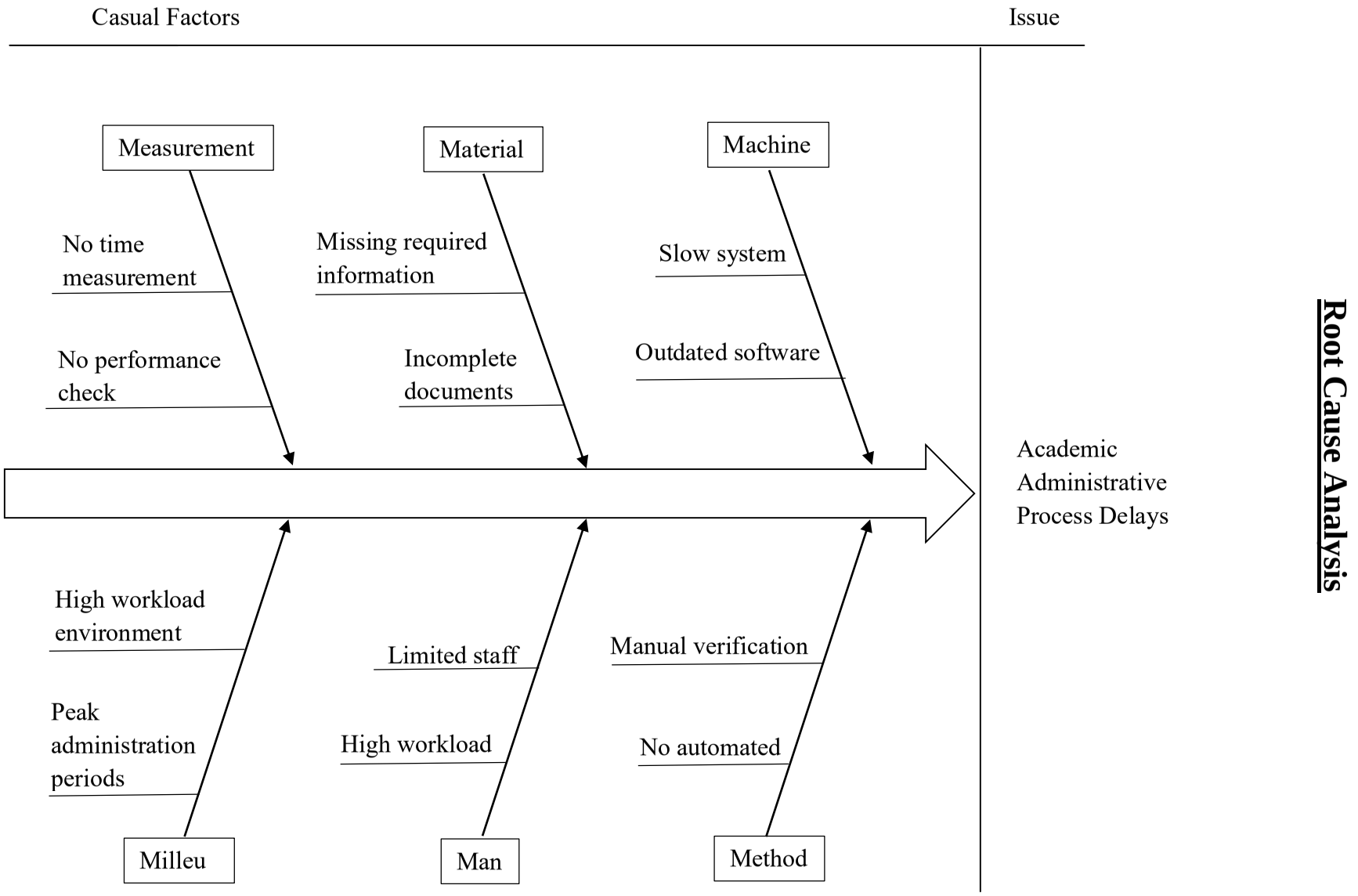
Assume each student loses 1 hour per week over 20 weeks.

Qualitative Impact:

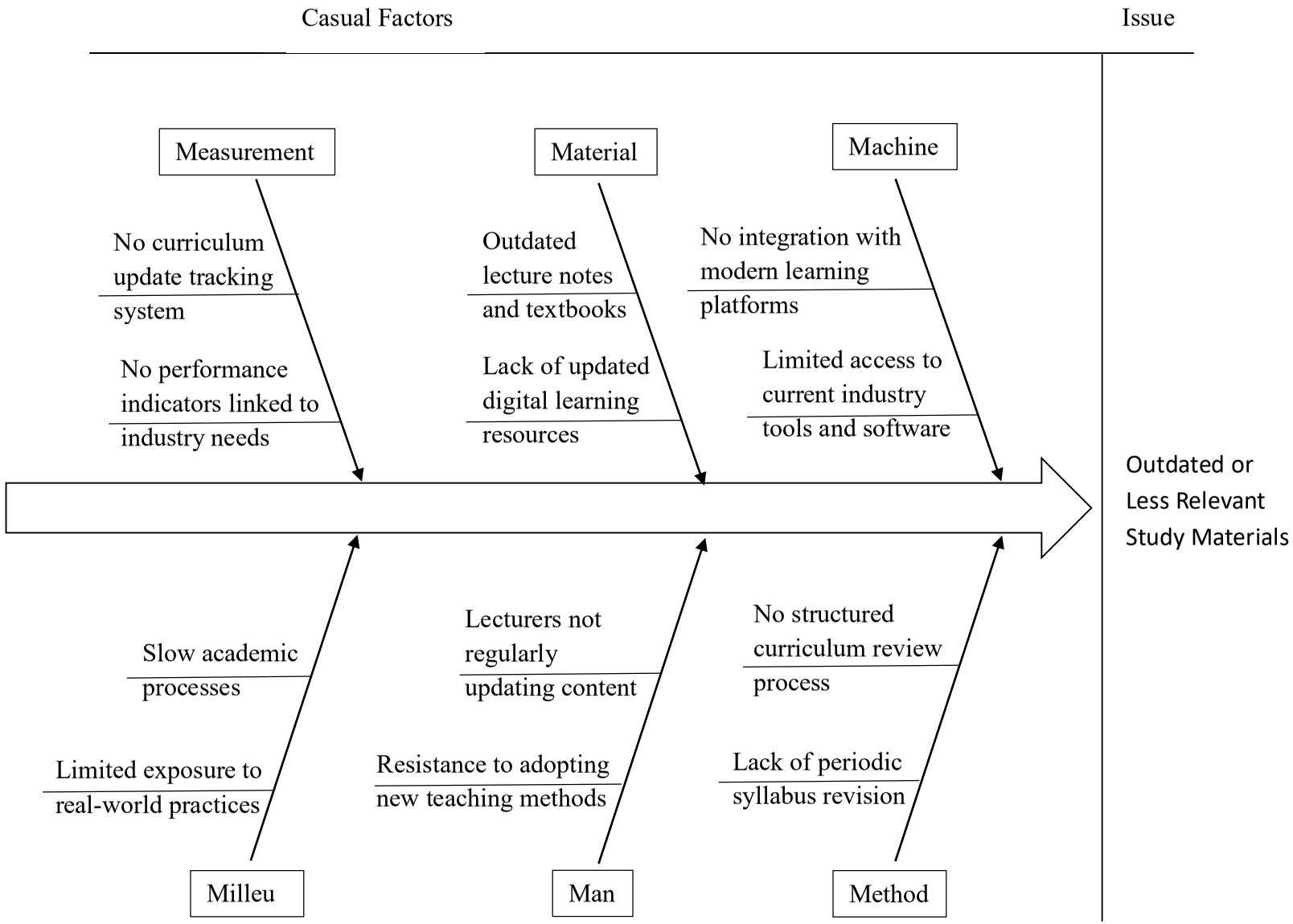
Reduces academic focus and overall performance.

Quantitative Impact:

1,000 × 1 × 20 = **20,000 hours wasted per year**



Root Cause Analysis


Casual Factors
Issue
Measurement
Material
Machine
No curriculum update tracking system
No performance indicators linked to industry needs
Outdated lecture notes and textbooks
Lack of updated digital learning resources
No integration with modern learning platforms
Limited access to current industry tools and software
Outdated or Less Relevant Study Materials
Slow academic processes
Limited exposure to real-world practices
Lecturers not regularly updating content
Resistance to adopting new teaching methods
No structured curriculum review process
Lack of periodic syllabus revision
Milleu
Man
Method

Process Implementation

Steps of Process Implementation

1. Student Application and Admission

    1. Identify the automation boundaries

        - Online application submission → automated
        - Document validation (basic checks) → automated
        - Merit evaluation (Z-score/criteria) → semi-automated
        - Final selection committee decision → manual
        - Admission letter generation → automated

    2. Review manual tasks

        - Interview evaluation (if required)
        - Final approval by admission officers
        - Special case handling (appeals, missing docs)

    3. Complete the process model

        - Data: student details, qualifications
        - Decision points: eligible / not eligible
        - Notifications: acceptance/rejection emails

    4. Adequate level of granularity

        - Submit application
        - Validate documents
        - Evaluate eligibility
        - Approve admission
        - Issue letter

    5. Specify execution properties

        - Variables: studentID, Z-score, eligibility status
        - Rules: admission criteria (cut-off marks)
        - Tasks:
            - Service task → eligibility check
            - User task → officer approval
            - Message → send admission result

2. Student Registration and Enrollment

1. Automation boundaries

- Registration form submission → automated
- ID generation → automated
- Course selection → semi-automated
- Final enrollment approval → manual

2. Manual tasks

- Verification of student identity
- Special approvals (late registration)

3. Complete process model

- System login creation
- VLE access
- Fee/payment confirmation

4. Granularity

- Accept offer
- Register student
- Assign courses
- Generate ID
- Activate systems

5. Execution properties

- Variables: studentID, course list
- Data mapping: admission → registration system
- Tasks:
    - Service → create account
    - User → approve enrollment

3. Deliver Academic Programs and Services

    1. Automation boundaries

        - Timetable scheduling → semi-automated
        - LMS (VLE content delivery) → automated
        - Lecture delivery → manual
        - Resource access → automated

    2. Manual tasks

        - Teaching lectures
        - Conducting labs
        - Project supervision

    3. Complete process model

        - Course materials
        - Attendance tracking
        - Feedback collection

    4. Granularity

        - Plan timetable
        - Conduct lectures
        - Provide materials
        - Manage labs
        - Facilitate projects

    5. Execution properties

        - Variables: courseID, student attendance
        - Systems: LMS, timetable system
        - Tasks:
            - User → teaching
            - Service → upload materials

4. Student Assessment and Evaluation

1. Automation boundaries

- Exam scheduling → automated
- Online exams → automated
- Assignment submission → automated
- Grading → manual/semi-automated

2. Manual tasks

- Marking papers
- Evaluating projects
- Moderation of marks

3. Complete process model

- Continuous assessment
- Exam evaluation
- Result entry

4. Granularity

- Conduct CA
- Conduct exams
- Evaluate answers
- Record marks

5. Execution properties

- Variables: marks, grades
- Rules: grading scheme
- Tasks:
    - User → marking
    - Service → calculate grade

5. Student Support and Development

1. Automation boundaries

- Career portal → automated
- Event registration → automated
- Counseling → manual

2. Manual tasks

- Mentoring sessions
- Workshops
- Career guidance

3. Complete process model

- Internship matching
- Event management
- Counseling services

4. Granularity

- Identify need
- Provide support
- Conduct workshops
- Track participation

5. Execution properties

- Variables: student skills, participation
- Tasks:
  - User → counseling
  - Service → internship matching

6. Research and Training

1. Automation boundaries

- Project registration → automated
- Progress tracking → semi-automated
- Internship placement → semi-automated
- Research supervision → manual

2. Manual tasks

- Assign supervisors
- Guide research
- Evaluate final research

3. Complete process model

- Proposal submission
- Progress reviews
- Final evaluation

4. Granularity

- Assign supervisor
- Conduct research
- Monitor progress
- Complete training

5. Execution properties

- Variables: research topic, progress status
- Tasks:
    - User → supervision
    - Service → progress tracking

7. Academic Results Processing

    1. Automation boundaries

        - Marks entry → semi-automated
        - GPA calculation → automated
        - Transcript generation → automated
        - Result approval → manual

    2. Manual tasks

        - Result verification
        - Approval by exam board

    3. Complete process model

        - Grade processing
        - Error checking
        - Result publishing

    4. Granularity

        - Enter marks
        - Calculate GPA
        - Verify results
        - Publish results

    5. Execution properties

        - Variables: GPA, grades
        - Rules: GPA formula
        - Tasks:
            - Service → GPA calculation
            - User → approval

8. Graduation and Completion

    1. Automation boundaries

        - Degree eligibility check → automated
        - Certificate generation → automated
        - Graduation ceremony → manual

    2. Manual tasks

        - Final approval
        - Ceremony organization

    3. Complete process model

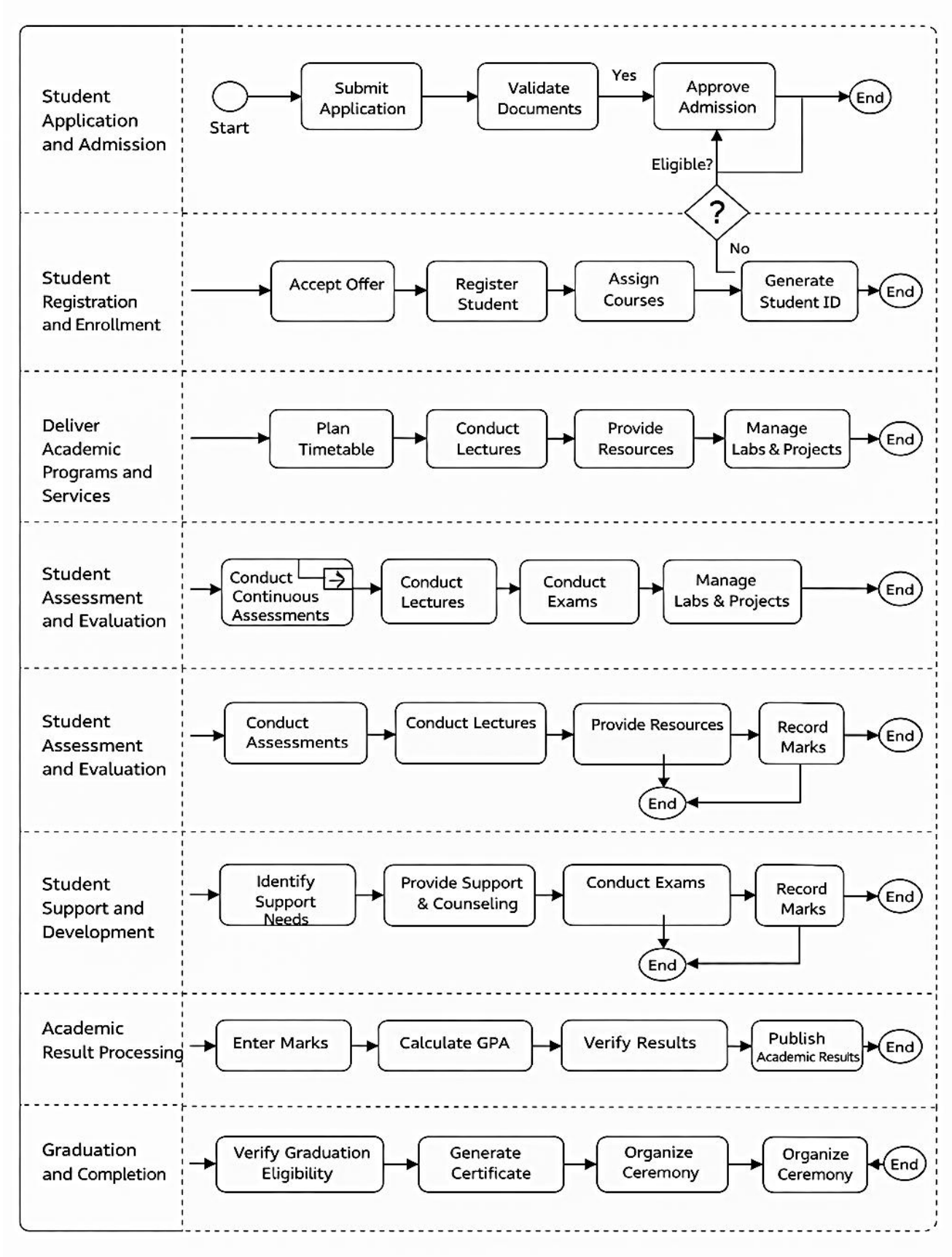
        - Eligibility verification
        - Certificate issuance
        - Alumni registration

    4. Granularity

        - Verify completion
        - Approve graduation
        - Issue certificate
        - Maintain alumni

    5. Execution properties

        - Variables: credits, GPA
        - Rules: graduation requirements
        - Tasks:
            - Service → eligibility check
            - User → final approval


Student Application and Admission
Start
Submit Application
Validate Documents
Yes
Approve Admission
End
Eligible?
?
No
Student Registration and Enrollment
Accept Offer
Register Student
Assign Courses
Generate Student ID
End
Deliver Academic Programs and Services
Plan Timetable
Conduct Lectures
Provide Resources
Manage Labs & Projects
End
Student Assessment and Evaluation
Conduct Continuous Assessments
Conduct Lectures
Conduct Exams
Manage Labs & Projects
End
Student Assessment and Evaluation
Conduct Assessments
Conduct Lectures
Provide Resources
Record Marks
End
End
Student Support and Development
Identify Support Needs
Provide Support & Counseling
Conduct Exams
Record Marks
End
End
Academic Result Processing
Enter Marks
Calculate GPA
Verify Results
Publish Academic Results
End
Graduation and Completion
Verify Graduation Eligibility
Generate Certificate
Organize Ceremony
Organize Ceremony
End

Process Monitoring

1. Student Application and Admission

Monitoring Focus

- Track how efficiently students are selected and admitted

Key Performance Indicators (KPIs)

- Application processing time
- Number of qualified applicants admitted
- Admission decision turnaround time

Monitoring Methods

- Operational Dashboard
- Daily application count, pending applications
- Monthly admission rates, rejection vs acceptance

Process Mining Use

- Analyze delays in evaluation stages
- Detect bottlenecks (e.g., document verification)

2. Student Registration and Enrollment

Monitoring Focus

- Ensure smooth and fast onboarding

KPIs

- Registration processing time
- System access success rate
- Enrollment completion rate

Monitoring Methods

- Dashboard showing:
- Number of students registered per day
- System errors/issues

Process Mining

- Identify delays in ID creation or system access
- Detect rework loops (students repeating steps)

3. Deliver Academic Programs and Services

Monitoring Focus

- Quality of teaching and learning delivery

KPIs

- Course completion rate
- Student satisfaction
- Teaching quality evaluation

Monitoring Methods

- Operational
- Attendance, lecture completion
- Tactical
- Semester performance reports
- Strategic
- Graduate skill development level

Process Mining

- Analyze delays in lectures, labs, and materials delivery
- Identify deviations from academic schedule

4. Student Assessment and Evaluation

Monitoring Focus

- Accuracy and timeliness of assessments

KPIs

- Grading accuracy
- Evaluation time
- Feedback turnaround time

Monitoring Methods

- Dashboard:
- Assignment submission vs grading completion
- Alerts for delayed marking

Process Mining

- Detect delays between exam and grading
- Identify inconsistencies in evaluation flow

5. Student Support and Development

Monitoring Focus

- Student engagement and development

KPIs

- Student participation rate
- Internship placement rate
- Student satisfaction

Monitoring Methods

- Dashboard tracking:
- Workshop attendance
- Counseling usage

Process Mining

- Analyze usage patterns of support services
- Identify low-engagement areas.

6. Research and Training

Monitoring Focus

- Research productivity and internship success

KPIs

- Research completion rate
- Internship success rate
- Number of publications

Monitoring Methods

- Dashboard:
- Research progress tracking
- Internship placement stats

Process Mining

- Detect delays in research milestones
- Analyze internship workflow efficiency

7. Academic Results Processing

Monitoring Focus

- Accuracy and speed of result processing

KPIs

- Result processing time
- Accuracy of results
- Transcript issuance time

Monitoring Methods

- Dashboard:
- Pending results
- Published results

Process Mining

- Detect bottlenecks in grading → result publishing
- Identify errors or rework cases

8. Graduation and Completion

Monitoring Focus

- Timely graduation and certification

KPIs

- Graduation processing time
- Degree verification accuracy
- Alumni engagement rate

Monitoring Methods

- Dashboard:
- Students eligible for graduation
- Degree issuance status

Process Mining

- Detect delays in eligibility verification
- Analyze graduation workflow completion time

Process Redesign

1. Student Application and Admission

Problems

- Delays due to manual document handling
- Repeated submissions
- Slow communication

Redesign

- Automation: Introduce full online admission system
- Integration: Connect UGC data directly to system
- Simplification: Remove duplicate document submission
- Parallelism: Process applications and eligibility checks at same time

Improved Outcome

- Faster admission decisions
- Reduced waiting time
- Better student experience

2. Student Registration and Enrollment

Problems

- Manual approvals
- System delays
- Slow access to university services

Redesign

- Automation: Auto-generate student ID, LMS access
- Integration: Link admission system → registration system
- Standardization: Use predefined registration workflows
- Self-service: Students complete registration online

Improved Outcome

- Quick enrollment
- Reduced administrative workload
- Less errors

3. Deliver Academic Programs and Services

Problems

- Outdated content
- Limited practical exposure
- Inefficient teaching methods

Redesign

- Task elimination: Remove outdated syllabus content
- Enhancement: Add industry-based projects
- Parallelism: Conduct theory + practical sessions together
- Technology use: Use LMS, recorded lectures, digital resources

Improved Outcome

- High-quality education
- Better skill development
- Industry-ready graduates

4. Student Assessment and Evaluation

Problems

- Delayed grading
- Manual evaluation
- Lack of fast feedback

Redesign

- Automation: Use online exams & auto-marking systems
- Parallelism: Evaluate assignments simultaneously by multiple lecturers
- Standardization: Use clear marking schemes
- Feedback improvement: Provide instant feedback via LMS

Improved Outcome

- Faster results
- Accurate grading
- Better student satisfaction

5. Student Support and Development

Problems

- Poor communication
- Low coordination
- Limited career guidance

Redesign

- Integration: Centralized student support platform
- Automation: Notifications for events, internships
- Enhancement: Introduce mentoring system
- Customer-focus heuristic: Focus on student needs

Improved Outcome

- Better student engagement
- Improved career opportunities
- Higher satisfaction

6. Research and Training

Problems

- Limited practical exposure
- Weak industry connection
- Poor supervision in internships

Redesign

- Integration: Connect university with industry partners
- Enhancement: Structured internship programs
- Parallelism: Combine research + industry projects
- Resource optimization: Assign supervisors efficiently

Improved Outcome

- Better research quality
- Strong industry exposure
- Higher employability

7. Academic Results Processing

Problems

- Delays in result processing
- Manual work
- Errors in grading

Redesign

- Automation: Use result processing systems
- Integration: Connect LMS + exam system
- Error reduction: Automatic grade calculation
- Simplification: Reduce approval layers

Improved Outcome

- Faster result release
- High accuracy
- Transparency

8. Graduation and Completion

Problems

- Delays in verification
- Manual processes
- Poor alumni engagement

Redesign

- Automation: Auto-check graduation requirements
- Integration: Connect results + degree system
- Simplification: Reduce manual verification
- Enhancement: Digital certificates & alumni system

Improved Outcome

- Faster graduation
- Accurate certification

Strong alumni network